

METHODOLOGY (AN OVERVIEW)

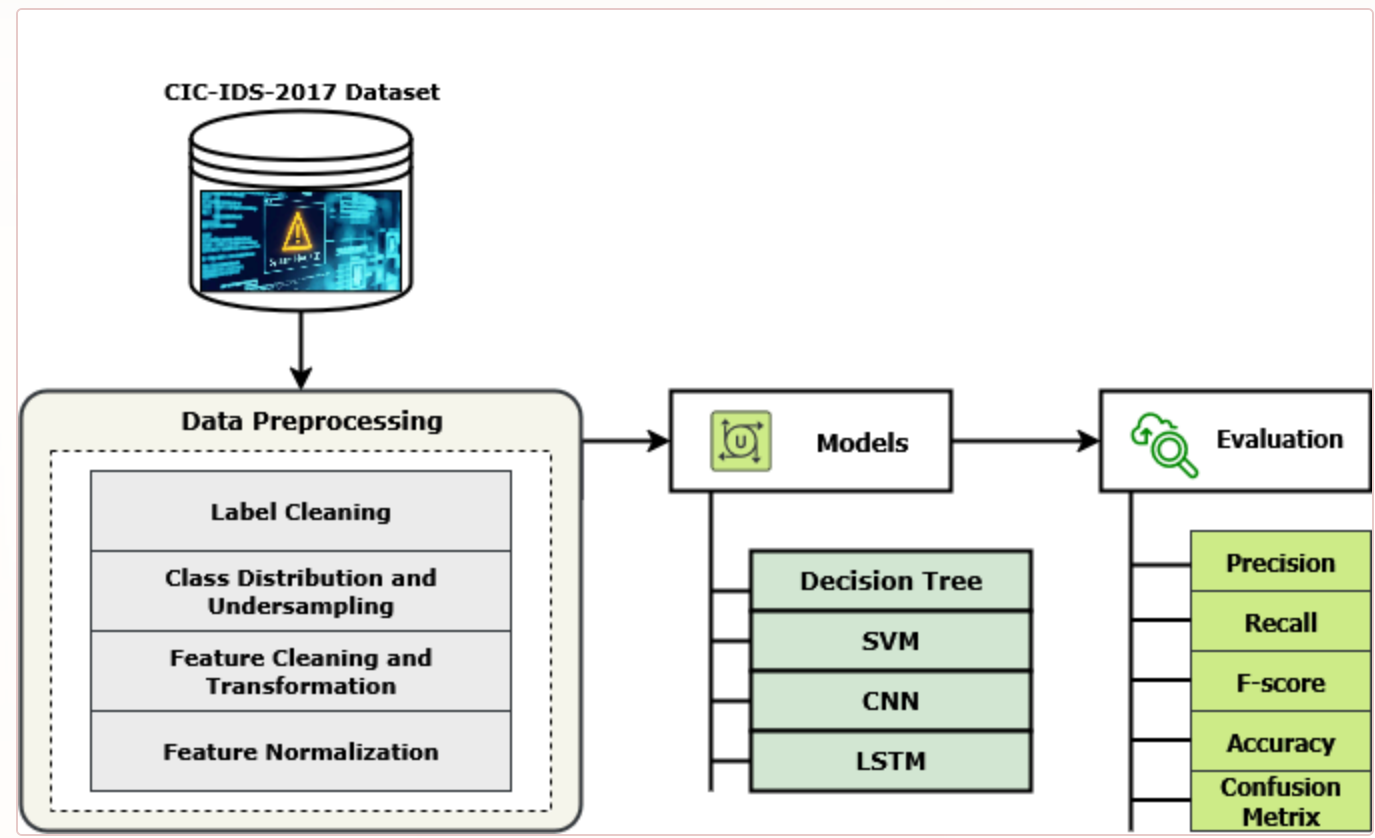
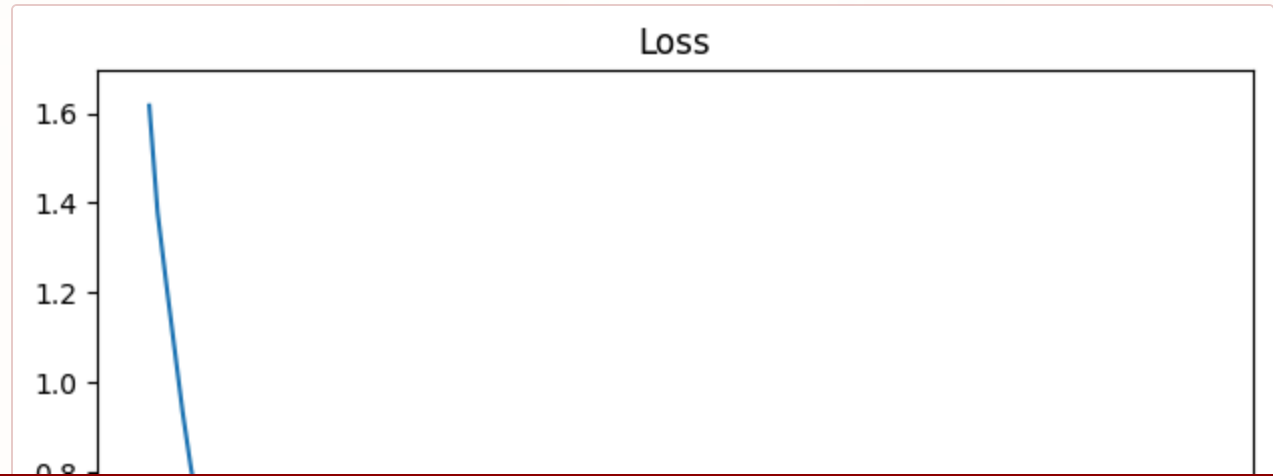
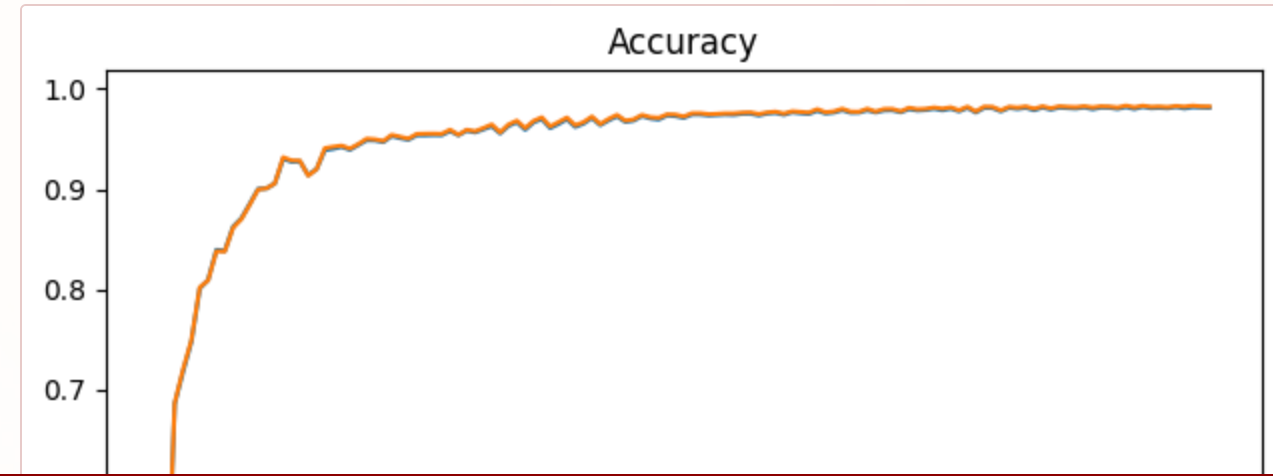


Fig. 1 – Serverless intelligent firewall system architecture

LSTM Parameter	Value
Input Features	78 Numerical flow features
Hidden units	3 layers: 128 → 64 → 32
Dropout	0.3 after each LSTM layer
Optimizer / LR	Adam / 0.001
Epochs / Patience	120 epochs - Early stop = 10
Lambda latency	<15 ms warm - <100 ms cold



RESULTS & EVALUATION

	98%	98%	98%	98%
SVM	84.1%	77.8%	80.8%	84.4%
Dec-Tree	87.6%	81.3%	84.2%	86.2%
CNN	95.1%	85.4%	89.9%	91.6%
LSTM	98.0%	98.0%	98.0%	98.0%

Model	Prec.	Recall	F1
SVM	84.1%	77.8%	80.8%
Dec-Tree	87.6%	81.3%	84.2%
CNN	95.1%	85.4%	89.9%
LSTM	98.0%	98.0%	98.0%

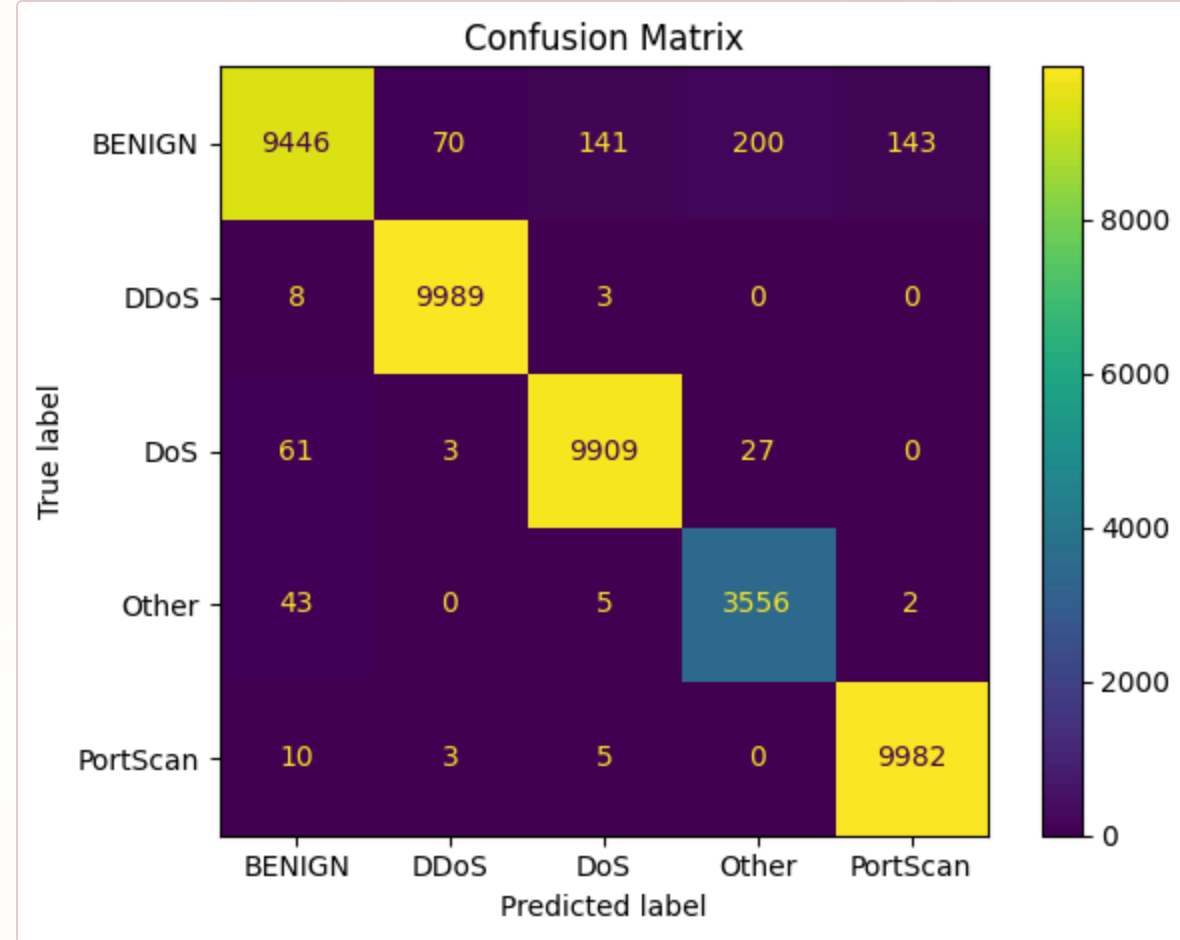


Fig. 4 – Confusion Matrix

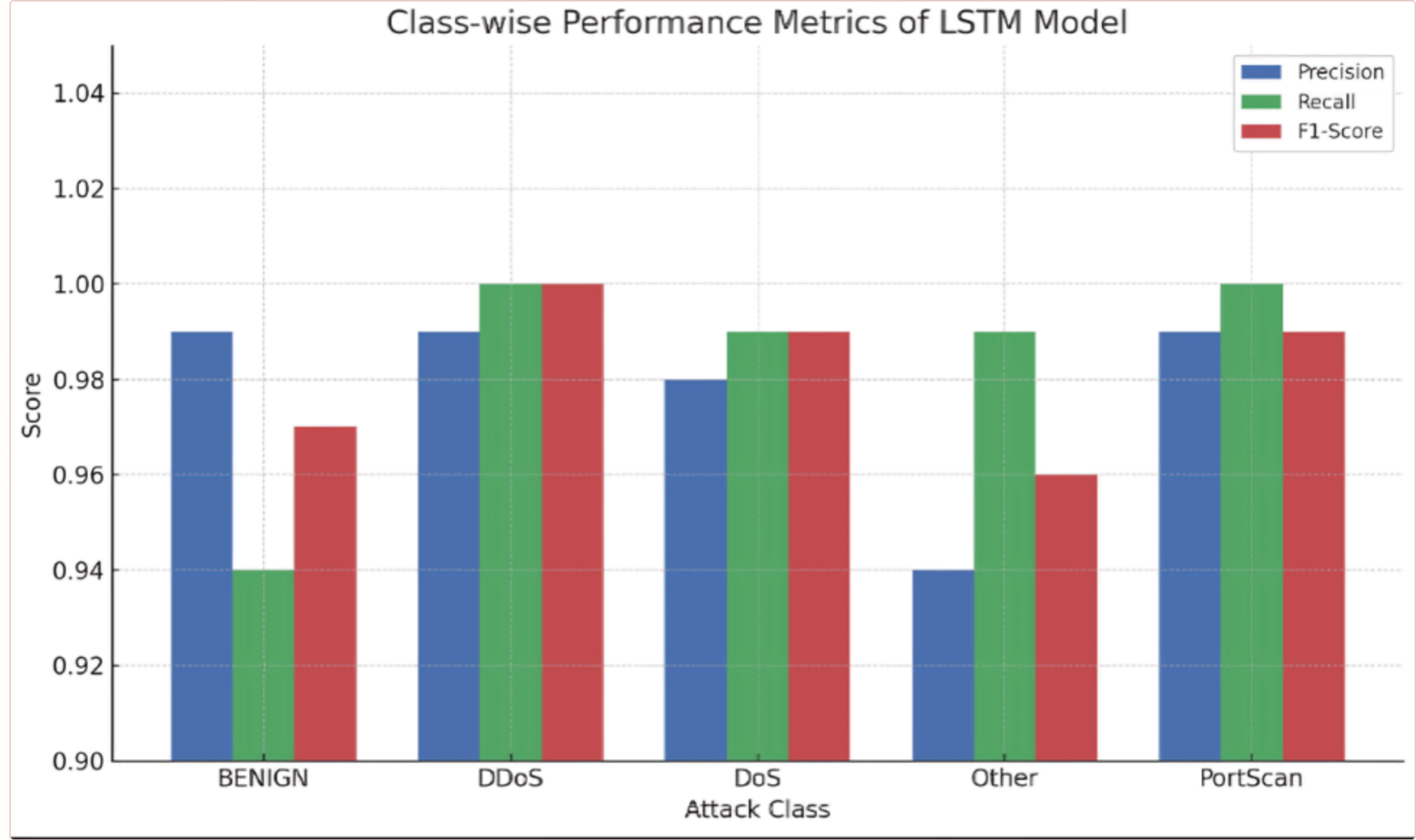


Fig. 5 – Class-wise Performance

Class	Precision	Recall	F1	Correct/Total
BENIGN	0.98	0.98	0.98	98/100